



International Material Data System

IMDS Recommendation	IMDS 027
----------------------------	-----------------

Rules and Guidelines for Reporting of Product Carbon Footprint (PCF)

1.	Disclaimer	2
2.	Purpose	2
3.	References	2
4	Definitions	4
4.1	Exempted Emissions in Catena-X PCF Rulebook	4
4.2	Production Site	4
5	PCF calculation	4
6	Data entry into IMDS	4
6.1	General	4
6.1.1	Transparency and visibility of the PCF data set	4
6.1.2	Product Carbon Footprint and required data versioning	5
6.1.3	Handling of standard / published MDS	5
6.2	Data entry into Ingredient Section – Product Carbon Footprint (PCF).....	6
6.2.1	Production site	6
6.2.2	Standards and Verification.....	7
6.2.3	Total PCF.....	8
6.2.4	Time Period.....	9
6.2.5	Exempted emissions.....	10
6.2.6	Primary Data Share (PDS)	10
6.2.7	Data Quality Rating (DQR)	10
6.2.8	Detailed GHG emissions	11
6.2.9	Carbon Content.....	12
6.3	Data entry into Recipient Section – Transport Carbon Footprint	12
6.3.1	Production Site.....	12
6.3.2	Customer Site	13
6.3.3	Coverage	13
6.3.4	Transport Carbon Footprint (TCF)	13
6.3.5	Detailed GHG emissions	13
7	Glossary	15
8	Release and Revisions.....	20
8.1	Release	20
8.2	Revision.....	20



International Material Data System

IMDS Recommendation**IMDS 027****1. Disclaimer**

Reporting of PCF data is **optional / not mandatory** per IMDS requirements, but there might be (contractual) B2B requirements, which mandate the reporting of PCF in IMDS. This is to be aligned with the direct customer.

In case, reporting is done in IMDS, the rules and guidelines of this document apply, and all data fields deemed mandatory are to be filled.

2. Purpose

This standard describes the general requirements for entering Product Carbon Footprint Data into IMDS and referencing to the Catena-X PCF Rulebook, which describes the correct calculation of this data.

Information about carbon footprints of the supplied materials and components is crucial for the whole automotive industry to demonstrate environmental responsibility, comply with regulations, meet consumer expectations, and stay competitive in an evolving marketplace focused on sustainability.

The IMDS Steering committee has decided to enhance IMDS to be used as a reporting tool for Product Carbon Footprints, to enable streamlined reporting based on the already established communications between supplier and customers, which are the send MDS. The infrastructure is provided with Release 15.

It has been decided to follow the Catena-X PCF Rulebook as an automotive standard for PCF calculations. Catena-X (CX) engages with respective automotive supplying sectors aligning the sectoral calculation methodologies. This process ensures compatibility of PCF reporting.

When external references change, e.g. the Catena-X PCF Rulebook it will be necessary for IMDS to review the impact and update the PCF reporting accordingly.

Changed scope between versions to be assessed and impacts on IMDS' PCF implementation.

Depending on extend of changes related IMDS enhancement proposals must be prepared and planned as part of an IMDS release and an update of Rec027 accordingly.

3. References**IMDS Recommendations**

IMDS 001 - General rules and guidelines for IMDS material data sheets

IMDS 025 - Evaluate the recycled and bio-based content of materials

PCF Methodology

ISO14067 Greenhouse gases — Carbon footprint of products — Requirements and guidelines for quantification



International Material Data System

IMDS Recommendation

IMDS 027

Overall automotive PCF rules

CX-0029 Product Carbon Footprint Rulebook, V. 3

<https://catenax-ev.github.io/docs/Jupiter/standards/CX-0029-ProductCarbonFootprintRulebook>

CX-0026 Product Carbon Footprint Data Model in line with referenced CX-0029 Rulebook

Catena-X Guidance

CX-Guidance Product Carbon Footprint (PCF) Calculation

Catena-X and TfS PCF Verification Guideline v.1.0

CX standards library

[Overview Standards | Catena-X - Library \(catenax-ev.github.io\)](https://catenax-ev.github.io)

CX training and support materials

<https://catena-x.academy>

Automotive supplier association guidance

AIAG's "A Simple Guide to Product Carbon Footprint"

[Product Carbon Footprint \(PCF\) in Automotive | AIAG](#)

PCF rules for sectors supplying automotive (aligned with CX)

TfS The Product Carbon Footprint Guideline for the Chemical Industry V.3

[Scope 3 GHG emissions programme - TfS Initiative \(tfs-initiative.com\)](#)

Worldsteel LCI methodology (2017)

<https://worldsteel.org/wp-content/uploads/Life-cycle-inventory-methodology-report.pdf>

European Aluminum methodology (2023)

<https://mailchi.mp/european-aluminium/l8webu401b>

International Aluminum Institute methodology (2021)

<https://international-aluminium.org/resource/aluminium-carbon-footprint-methodology/>



International Material Data System

IMDS Recommendation**IMDS 027**

4 Definitions

4.1 Exempted Emissions in Catena-X PCF Rulebook

These are named Cut-Off Rules in the Catena-X PCF Rulebook.

4.2 Production Site

This represents the site for which the PCF was calculated for and applies, which is the production site of the product manufacturer or the warehouse / logistic center of the distributor / wholesaler. It might be equal to the shipping site.

In case of multi-site production, this information can be set to country, regional or even global level.

5 PCF calculation

It has been decided to follow the Catena-X PCF Rulebook as an automotive standard for PCF calculations. Catena-X engages with respective automotive supplying sectors aligning the sectoral calculation methodologies. This process ensures compatibility of PCF reporting.

PCF data will only need to be calculated based on the Catena-X PCF Rule Book or other accepted rulebooks/standards (see Standards and Verification) and the data can then be reported also into other tools/systems (like Catena-X) that your automotive customer requires you to use.

For those suppliers to other industry sectors where additional/different rulebooks may need to be used, please check for cross-recognition. If the Catena-X PCF Rulebook is cross-recognized, existing data will be accepted. Please contact your customer for additional clarification, if needed.

6 Data entry into IMDS

6.1 General

This section describes the input of PCF data into IMDS. Information in this section details the mandatory fields that must be completed and the rules governing their input.

You will not find any calculation information in this section as this must be completed outside of the IMDS system using your preferred tool and mainly the Catena-X PCF Rulebook.

If other aligned rulebooks or standards are used for calculation, there are certain additional requirements described by Catena-X.

6.1.1 Transparency and visibility of the PCF data set

Transparency of the PCF data set along the supply chain:

The visibility of PCF data reporting follows the 'one-level up/one-down' rule for the time being.

- a. The PCF data set will be visible only for the direct customer.
- b. Data Quality Rating (DQR) and Primary Data Share (PDS) visibility throughout the supply chain.

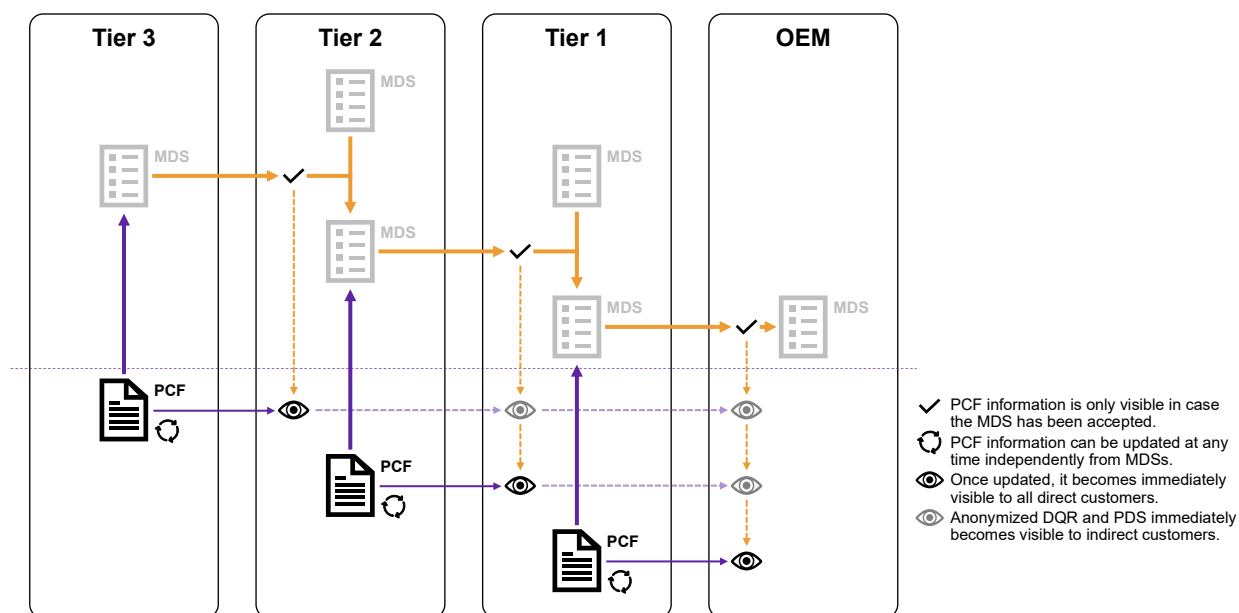
If there is no transport carbon footprint (TCF) defined, DQR and PDS are visible for all entered production sites.

International Material Data System

IMDS Recommendation

IMDS 027

- c. PCF data set reported via IMDS is visible only after the respective MDS is accepted.



6.1.2 Product Carbon Footprint and required data versioning

Product Carbon Footprint (PCF) data in IMDS has separate versioning from the actual MDS to which it is attached so that the information can be revised without requiring resubmission.

Rule/Guideline	Description
Rule 6.1.2.A	Data must be reviewed at least once per year and mandatorily updated when an increase by 5 percent occurs in the reported values. The detailed update rules of the Catena-X PCF Rulebook apply, deviating customer requirements might exist.
Rule 6.1.2.B	Structural changes in the operation shall lead to a mandatory update as well.
Rule 6.1.2.C	Changes in the calculation (like methodology, data accuracy) leading to significant impact shall also lead to a mandatory update. Small changes may accumulate to be significant.
Rule 6.1.2.D	Significant errors shall also lead to a mandatory update. Small errors may accumulate to be significant.

The enhanced reporting allows companies to provide PCF along the supply chain in a faster way without blocking or interfering in the "classic" MDS data traffic and still maintaining the link between the added regulatory information and the related MDSs. This allows for immediate updating of PCF information as changes occur in the product.

6.1.3 Handling of standard / published MDS

- **Standard Material MDS handling:**
Standard Material MDS do not contain PCF information and Users cannot enter it on this level. PCF information (calculated according to an accepted rulebook) of those materials must be

International Material Data System

IMDS Recommendation

IMDS 027

requested from the real material manufacturer outside of IMDS and incorporated into the PCF calculation of the referencing (upper level) MDS in line with IMDS001 Recommendation.

- Published MDS handling (excluding published Standard MDSs):**
 Entering PCF data is enabled for owners of published MDSs (materials, semi-components, and components).
 All production sites will be visible for published MDSs so the user can select the correct production site and accordingly include the applied transport CF in the calculation (outside of IMDS).

6.2 Data entry into Ingredient Section – Product Carbon Footprint (PCF)

The following section describes the requirements of the *Ingredient Section* in case PCF information is entered into IMDS.

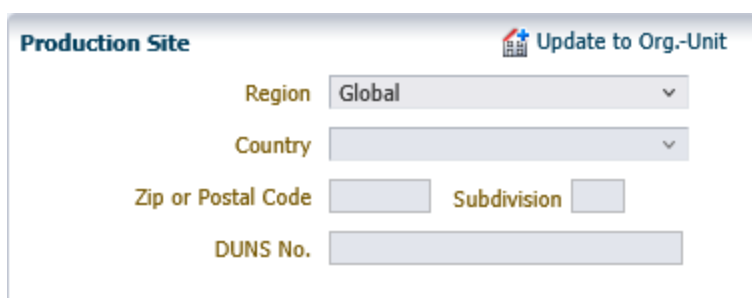
There are several fields to be entered when providing PCF data in the *Ingredient Section*. Each field provides a certain kind of information. Some of this information is mandatory, other is optional. This section describes each field and the respective rules.

6.2.1 Production site

See chapter 4.2 Definition – Production Site.

The field *Region* is set to “Global” by default. Clicking *Update to Org.-Unit* automatically fills the fields of *Region*, *Country*, *ZIP Code* and *DUNS-No.* based on the *Org.-Unit* assigned to the MDS. When another *Region* is manually selected, the *Country* field becomes available. Only when a *Country* is selected, *Subdivision*, *Zip Code* and *DUNS No.* become available. After release of PCF information, existing values except for *DUNS No.* cannot be changed.

While new versions of PCF information may contain additional production site details, other details part of previous versions may not be changed. E.g., a *Country* can be specified if it was not included in the last version but may not be changed again in later versions. To declare PCF information for another *Country*, a new *Production Site* must be created.



Rule/Guideline	Description
Rule 6.2.1.A	At least the <i>Region</i> must be selected in case of PCF information.
Rule 6.2.1.B	The <i>Production Site</i> information should be selected as used for the calculation. To address multi-site production regional or global production sites can be selected.
Rule 6.2.1.C	The <i>Org.-Unit</i> must represent the correct production site if <i>Update to Org.-Unit</i> is used.

International Material Data System

IMDS Recommendation

IMDS 027

Guideline 6.2.1.a	Other than <i>Region</i> the entry of <i>Subdivision</i> is optional. <i>Subdivision</i> is a free text field for easier maintenance and limited to the same number of characters as in the Catena-X PCF Rulebook (2 characters).
-------------------	---

6.2.2 Standards and Verification

Standards / Rulebook:

The standard / rulebook, which was used for the PCF calculation must be selected. PCF calculation for materials can be done based on selected rulebooks other than the latest Catena-X PCF Rulebook. Anyhow, while using those selected rulebooks for calculation, specific alignments and additional requirements must be considered. Those alignments are documented in the annex of the Catena-X PCF Rulebook.

Standards and Verification

Standard / Rulebook
?

Certification / Verification
done by appointed 3rd party
?

Further details / link

Rule/Guideline	Description
Rule 6.2.2.A	The <i>Standard / Rulebook</i> used for the calculation of the PCF must be selected in case of PCF registration. With the Catena-X PCF Rulebook v3 the following Standards / Rulebooks are defined: - Catena-X v3 (2024) - TfS V3.0 (2022) + additional req. - worldsteel (2017) + additional req. - IAI v2.0 (2021) + additional req. - EA Rev.8 (2023) + additional req. PCF information calculated based on other rulebooks are not accepted.
Rule 6.2.2.B	The only viable rulebook to be selected for components is the Catena-X PCF Rulebook v3. This rulebook can also be used for semi-components and materials. This might be changed in the future. All other included rulebooks are to be used exclusively for the respective semi-components and materials – current scope: TfS for chemicals, worldsteel for steel, IAI and EA for aluminum and its alloys.

Certification / Verification done by appointed 3rd party

IMDS allows the selection of the *Certification / Verification* methodology used to ensure the quality of the reported PCF. The logic is in line with the Catena-X PCF Rulebook and TfS PCF Verification Framework V1.

Verification refers to the process of evaluating PCF data set to determine whether it complies with the relevant standard.

International Material Data System

IMDS Recommendation

IMDS 027

Standards and Verification

Standard / Rulebook

Certification / Verification done by appointed 3rd party

Further details / link

Rule/Guideline	Description
Rule 6.2.2.C	The <i>Certification / Verification</i> must be done based on the Catena-X PCF Rulebook and TfS PCF Verification Framework. Currently the drop-down list allows three different selections, beside “No” for non-certified/verified PCF data: - “Yes, PCF program certification (3rd party) + any level of verification” - “Yes, product family verification” - “Yes, part number specific verification” The respective free text field becomes available in case of any of the “Yes” statements and must be used.
Guideline 6.2.2.a	If “Yes” for <i>Certification / Verification</i> has been selected (Rule 6.2.2.C) additional information must be entered: 1. Weblink to the related certificate / verification document, which is publicly accessible, and it is ensured to be accessible at least until a new version of the PCF information with a new certificate / verification document is available. - or 2. Further information, which allow the access to the certificate / verification document, like certification body and certificate number, if this certification body allows access to the respective certificate with the certificate number. - or 3. If the named PCF contact is also responsible for the certification/verification it should be added.
Rule 6.2.2.E	In case of a 2nd party verification (customer verification), the identity of this party must not be disclosed in the information (text or weblink) given in the free text field. The identity of the 2nd party verifier must be anonymized.

6.2.3 Total PCF

The values are always attributed to the declared unit of a product by adding up the CO2 equivalents of all attributable processes along the cradle-to-gate of the product.

Total PCF includes all attributable upstream and direct emissions of producing a product, **including all upstream transportation activities**.

International Material Data System

IMDS Recommendation

IMDS 027

Product Carbon Footprint

PCF of referenced MDSs - ?

Total PCF (excluding biogenic) kg CO2e / component ? *

Total PCF (including biogenic) kg CO2e / component ? *

Time Period - ? *

Exempted emissions % ? *

Exempted emissions description ?

Rule/Guideline	Description
Rule 6.2.3.A	<i>Total PCF (including biogenic)</i> and <i>Total PCF (excluding biogenic)</i> are both mandatory. If no further details are known, they have the same value.
Rule 6.2.3.B	For components the unit is "kg CO2e / component". For semi-component(s) under semi-component, the PCF value is always given in "kg CO2e/material kg" for the main semi-component only. The percentages for the sub-semi-components are considered. For material(s) under material, the PCF value is always given in "kg CO2e/material kg" for the main material only. The percentages for the sub-materials are considered.
Guideline 6.2.3.a	<i>Total PCF (including biogenic)</i> very likely exceeds the value "PCF of Referenced MDSs" if PCF values of supplied sub-components/materials can be considered as correct PCF values.
Rule 6.2.3.C	The <i>Total PCF (excluding biogenic)</i> values can only contain positive values. The <i>Total PCF (including biogenic)</i> values can also contain negative values.

6.2.4 Time Period

The *Time Period* to be entered is the period the PCF values were calculated for/in. **It is not a validity time.**

Rule/Guideline	Description
Rule 6.2.4.A	The entry of <i>Time Period</i> is mandatory.
Guideline 6.2.4.a	Emissions shall be reported averaged over one year (reporting or calendar year) to reflect typical production conditions and avoid possible fluctuations. Shorter periods may be reported if emission average of a full year is not yet available (like production ramp-up phases). Longer periods may be reported but shall not exceed three years. Any period deviating from the one-year average default might need additional justification.

International Material Data System

IMDS Recommendation

IMDS 027

6.2.5 Exempted emissions

If the individual material or energy flow are found to be insignificant for the carbon footprint those can be documented as *Exempted Emissions*.

Rule/Guideline	Description
Rule 6.2.5.A	The entry of <i>Exempted Emissions</i> is mandatory and must be in the range of 0 - 3%.
Rule 6.2.5.B	Justification to be given in case the value is greater than zero.

6.2.6 Primary Data Share (PDS)

Primary Data Share represents the amount of primary data used for the calculation of the PCF.

The *PDS* is to be aggregated along the supply chain, taking into account the contribution to the Total PCF. The reference is the Total PCF (including biogenic), the aggregation is to be calculated outside of IMDS.



Rule/Guideline	Description
Rule 6.2.6.A	The entry of <i>Primary Data Share</i> is mandatory.
Rule 6.2.6.B	The PDS must be in the range 0% to 100%.
Rule 6.2.6.C	PDS must be set to 0 in case not existing in sectorial guidance. (see Catena-X Annexes).

6.2.7 Data Quality Rating (DQR)

Data Quality Rating is a self-assessment to indicate, how representative the values (Total PCF and others) are in connection to the declared product.

The DQR is not aggregated along the supply chain but only reported by the user for the processes and PCF calculation within its responsibility.

It is separated into three different categories technology, time, geography and the definition of those categories are to be found in the Catena-X PCF Rulebook and the Catena-X Annexes in case another rule book was used for calculation.

International Material Data System

IMDS Recommendation

IMDS 027

PDS / DQR

PDS % ? *

DQR

Technology (TeR) ? *

Time (TiR) ? *

Geography (GeR) ? *

Rule/Guideline	Description
Rule 6.2.7.A	The entry of <i>Data Quality Rating</i> is mandatory.
Rule 6.2.7.B	Each DQR must be in the range 1 (best) to 5 (worst).
Rule 6.2.7.C	DQR must be set to 5 in case not existing in sectorial guidance. (see Catena-X Annexes).

6.2.8 Detailed GHG emissions

The *detailed GHG emissions* are a subset of the Total PCF values, which might not be known. The default for each field is therefore zero.

To fill information about *detailed GHG emissions* (calculated based on Catena-X PCF Rulebook or Together for Sustainability PCF Guideline), open the optional fields and enter the required GHG emissions values.

Detailed GHG emissions ?

Net fossil GHG emissions and removals 0.0 kg CO2e / component *

Net biogenic GHG emissions and removals other than biogenic CO2 0.0 kg CO2e / component *

Net biogenic GHG emissions and removals including biogenic CO2 0.0 kg CO2e / component *

GHG emissions and removals from land use change 0.0 kg CO2e / component *

GHG emissions and removals from direct land use change 0.0 kg CO2e / component *

GHG emissions resulting from aircraft transportation 0.0 kg CO2e / component *

Rule/Guideline	Description
Rule 6.2.8.A	The entry of <i>Detailed GHG Emissions</i> is mandatory, however using the default value is allowed, if information is not available.
Guideline 6.2.8.a	If direct suppliers report values in this section, they should be reflected in the report of the final product. Manual check and calculation are required beforehand to ensure the data is correct.

International Material Data System

IMDS Recommendation

IMDS 027

6.2.9 Carbon Content

Carbon Content (calculated based on TfS PCF Guideline) can be entered by expanding the *Carbon Content* section. The default for each field is zero.

Carbon Content

0.0

kg C / component

?

*

Biogenic Carbon Content

0.0

kg C / component

?

*

Rule/Guideline	Description
Rule 6.2.9.A	The entry of <i>Carbon Content</i> is mandatory, however using the default value is allowed, if information is not available.
Guideline 6.2.9.a	If direct suppliers report values in this section, they should be reflected in the report of the final product. Manual check and calculation are required beforehand to ensure the data is correct.

6.3 Data entry into Recipient Section – Transport Carbon Footprint

Transport Carbon Footprint only includes GHG emissions from transport between supplier's outbound gate and a customer inbound gate.

Each logistic process prior to the supplier outbound gate is considered inhouse or upstream transportation and must therefore be included in the PCF calculation. It is not considered Transport Carbon Footprint.

Transport CF

Production Site

*

Customer Site

Coverage

Supplier responsibility

?

until customer inbound gate

full transport covered

Transport Carbon Footprint (excluding biogenic)

kg CO2e / component

?

*

Transport Carbon Footprint (including biogenic)

kg CO2e / component

?

*

6.3.1 Production Site

The appropriate *production site* must be selected, to report the corresponding transport carbon footprint. Those are the ones already entered for PCF creation.



International Material Data System

IMDS Recommendation

IMDS 027

6.3.2 Customer Site

Location or site at which control of the product is transferred to the customer's inbound gate. It can be the site that is listed on the PPAP PSW documentation.

This is a free text field description.

The *customer site* must be clearly identifiable.

6.3.3 Coverage

Description, which part of the transport is covered by the reported Transport Carbon Footprint.

Customer responsibility - means that customer is responsible for Transport Carbon Footprint information from the time they assume control of product = supplier's outbound gate. No information to be added into IMDS.

Supplier responsibility - means supplier responsible to report Transport Carbon Footprint data up until product control is handed over = customer's inbound gate. Information to be added by the supplier.

Split responsibility – means an agreement will determine which part of the transport is covered by the reported Transport Carbon Footprint data. In case of “split responsibility” communication outside of IMDS is necessary with the direct customer, to align the exact boundaries of the reported Transport Carbon Footprint. Information to be added by the supplier, *additional Transport carbon footprint to be included in the customer calculation*.

6.3.4 Transport Carbon Footprint (TCF)

This is the calculated number at the product level based on the transportation method used for the product.

Transport Carbon Footprint must be reported as average values.

Rule/Guideline	Description
Rule 6.3.4.A	<i>TCF (including biogenic)</i> and <i>TCF (excluding biogenic)</i> are both mandatory. If no further details are known, they have the same value.
Rule 6.3.4.B	The <i>TCF (excluding biogenic)</i> values can only contain positive values. The <i>TCF (including biogenic)</i> values can also contain negative values.

6.3.5 Detailed GHG emissions

The *detailed GHG emissions* are a subset of the TCF values, which might not be known. The default for each field is therefore zero.

To fill information about *detailed GHG emissions* (calculated based on Catena-X PCF Rulebook or Together for Sustainability PCF Guideline), open the optional fields and enter the required GHG emissions values.



International Material Data System

IMDS Recommendation

IMDS 027

▼ Detailed GHG emissions ?

Net fossil GHG emissions and removals	0.0	kg CO2e / component *
Net biogenic GHG emissions and removals other than biogenic CO2	0.0	kg CO2e / component *
Net biogenic GHG emissions and removals including biogenic CO2	0.0	kg CO2e / component *
GHG emissions and removals from land use change	0.0	kg CO2e / component *
GHG emissions and removals from direct land use change	0.0	kg CO2e / component *
GHG emissions resulting from aircraft transportation	0.0	kg CO2e / component *

Rule/Guideline	Description
Rule 6.3.5.A	The entry of <i>Detailed GHG Emissions</i> is mandatory, however using the default value is allowed, if information is not available.

International Material Data System

IMDS Recommendation

IMDS 027

7 Glossary

Term	Abbreviation	Definition	Source / Reference
Attributable process		Those processes that consists of all service, material and energy flows that become, make and carry a product throughout its life cycle.	WBCSD Pathfinder
Biogenic carbon		Carbon derived from biomass	ISO 14050:2020(en) Environmental management — Vocabulary, 3.8.24
carbon dioxide equivalent	CO ₂ e	Unit for comparing the radiative forcing of a greenhouse gas to that of carbon dioxide	ISO 14050:2020(en) Environmental management — Vocabulary, 3.9.3.
Cradle to gate		System boundary that is applied for a partial PCF assessment that includes a part of the product's life cycle. Cradle-to-gate represents the GHG emissions and removals arising from all life cycle stages, up to the point where the product leaves the production site (the "gate"). This explicitly excludes the life cycle stages use and end-of-life.	adapted from TfS PCF Guideline and in reference to ISO 14067 6.3.4.2 System boundary options
Cut-Off criteria		Specification of the amount of material or energy flow or the level of significance of GHG emissions associated with unit process or the product system, to be excluded from a PCF study	DIN EN ISO 14067, Feb. 2019, p. 24



International Material Data System

IMDS Recommendation

IMDS 027

Data Quality Rating	DQR	Qualitative rating to assess the representativeness of PCF data regarding the technology, time period, and geography of the declared product. DQR is composed of three sub-ratings: TeR, TiR, and GeR. Each is rated on a scale from 1 (best) to 5 (worst). The representativeness (technological, geographical, and time-related) characterizes the degree to which the processes and products selected depict the system analyzed	Catena-X PCF Rulebook
Data Universal Numbering System Number	DUNS	A unique identifier for business entities used in IMDS	
Declared Unit		Quantity of a product for use as a reference unit in the quantification of a Cradle-to-Gate PCF	adapted from DIN EN ISO 14067
Direct emissions		GHG emissions from the processes that are owned or controlled by the reporting company	WBCSD Pathfinder
Exempted Emissions		Applied cut off criteria, it specifies the amount of emissions excluded (cut off) from total emissions. Catena-X PCF Rule book allows max 3 % cut off (corresponding to at least 97% coverage of emissions)	
Geographical Representativeness	GeR	Assesses the extent to which the data represents the actual location or region where the process takes place.	Adapted from Catena-X PCF Rulebook
Greenhouse Gas	GHG	Gaseous constituent of the atmosphere, both natural and anthropogenic, that absorbs and emits radiation at specific wavelengths within the spectrum of infrared radiation emitted by the	DIN EN ISO 14067, Feb. 2019, p. 19



International Material Data System

IMDS Recommendation

IMDS 027

		Earth's surface, the atmosphere, and clouds.	
In Bound		Inbound logistics: All transportation activities related to receiving goods, materials, or components from external suppliers to the own organization for further processing or use in production.	
Out-bound gate		The supplier's factory gate, through which the product leaves the production site and is ready for shipment to the customer.	Adapted from Catena-X PCF Rulebook
Material Data Sheet	MDS	Data entries in IMDS for materials, semi-components, or components.	
Negative emissions		A situation of net negative emissions is achieved when, as the result of human activities, more green-house gases are removed from the atmosphere than are emitted into it. Where multiple greenhouse gases are involved, the quantification of negative emissions depends on the climate metric chosen to compare emissions of different gases (such as global warming potential, global temperature change potential, and others, as well as the chosen time horizon).	IPCC glossary
Net negative emissions		Removal of greenhouse gases (GHGs) from the atmosphere by deliberate human activities, i.e., in addition to the removal that would occur via natural carbon cycle processes	IPCC glossary
Org.-Unit		Organizational Unit – part of a company structure within IMDS, tied to a production site.	



International Material Data System

IMDS Recommendation

IMDS 027

Primary Data		Primary data is a quantified value of a process, or an activity obtained from a direct measurement, or a calculation based on direct measurements. Primary data can include greenhouse gas emission factors and/or greenhouse gas activity data. Average data from industry associations or global averages do not qualify as primary data.	14067:2018, 3.1.6.1
Primary Data Share	PDS	The Primary Data Share (PDS) is defined as the proportion (percentage) of a PCF that is derived from primary data. When calculating the primary data share, a contribution to the primary data share shall only be attributed, 953 if both the activity data (e.g., amount in kWh) and the GHG emission factor (expressed in kgCO ₂ e per unit) 954 information is derived from primary.	Adapted from Catena-X PCF Rulebook
(total) Product Carbon Footprint	PCF	Total GHG emissions generated during the life cycle of a product, measured in CO ₂ e. Within the boundary of the CX-PCF Rulebook, emissions related to the product use and end-of-life stages are excluded from the PCF. Including biogenic emissions.	WBCSD Pathfinder
(total) Product Carbon Footprint excluding biogenic emissions		Total GHG emissions generated during the life cycle of a product, measured in CO ₂ e. Within the boundary of the CX-PCF Rulebook, emissions related to the product use and end-of-life stages are excluded from the PCF.	



International Material Data System

IMDS Recommendation

IMDS 027

		excluding biogenic emissions.	
Technology Representativeness	TeR	Assesses how well the data reflects the specific technology used in the production / product of the declared product	Adapted from Catena-X PCF Rulebook
Temporal Representativeness	TiR	Describes how current the data is in relation to the reporting period.	Adapted from Catena-X PCF Rulebook
(total) Transport Carbon Footprint	TCF	Total GHG emission generated during logistic activities of a product (transport, transshipment, storage), including biogenic emissions	
(total) Transport Carbon Footprint excluding biogenic emissions		Total GHG emission generated during logistic activities of a product (transport, transshipment, storage), excluding biogenic emissions	
Upstream emission		Indirect GHG emissions that occur in the value chain prior to the processes owned or controlled by the reporting company. All upstream transportation emissions are also included as part of upstream emissions.	WBCSD Pathfinder

As of April 16, 2025



International Material Data System

IMDS Recommendation**IMDS 027****8 Release and Revisions****8.1 Release**

The recommendation was first approved and released in 17/06/2025

8.2 Revision

Rev.	Date	Description / Reason	Originating Organization / Committee
1.0	17/06/2025	Approval received from AIAG, CLEPA, JAPIA and IMDS SC.	IMDS 027 PCF WG
	02/07/2025	Updated changed CX PCF Rulebook web link	DXC
	29/10/2025	Corrected typos in 6.2.3 B and 6.3	DXC
	15/06/2026	Update of chapter 3 References with AIAG guidance document link as agreed in IMDS SC269 Supplier joint meeting	DXC